generate_UCH_metagenomics_plots

2026-03-05

```
library(ggplot2)
```

```
## Warning: package 'ggplot2' was built under R version 4.4.3
```

```
library(stringr)
library(plyr)
library(dplyr)
```

```
##
## Attaching package: 'dplyr'
```

```
## The following objects are masked from 'package:plyr':
##
##     arrange, count, desc, failwith, id, mutate, rename, summarise,
##     summarize
```

```
## The following objects are masked from 'package:stats':
##
##     filter, lag
```

```
## The following objects are masked from 'package:base':
##
##     intersect, setdiff, setequal, union
```

```
library(data.table)
```

```
##
## Attaching package: 'data.table'
```

```
## The following objects are masked from 'package:dplyr':
##
##     between, first, last
```

```
library(cowplot)
library("FactoMineR")
library(pheatmap)
library(vegan)
```

```
## Warning: package 'vegan' was built under R version 4.4.2
```

```
## Loading required package: permute
```

```
## Warning: package 'permute' was built under R version 4.4.2
```

```
## Loading required package: lattice
```

```
## This is vegan 2.6-8
```

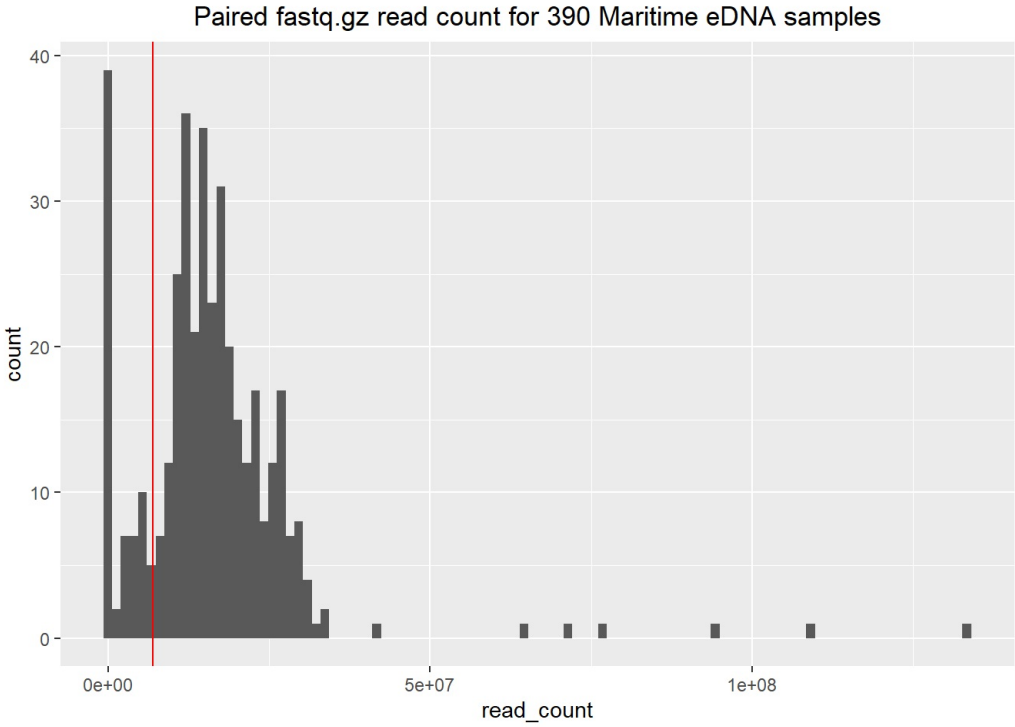
```
library(readxl)
library(tidyr)
library(pathview)
```

```
##
```

```
## ##############################################################################
## Pathview is an open source software package distributed under GNU General
## Public License version 3 (GPLv3). Details of GPLv3 is available at
## http://www.gnu.org/licenses/gpl-3.0.html. Particullary, users are required to
## formally cite the original Pathview paper (not just mention it) in publications
## or products. For details, do citation("pathview") within R.
##
## The pathview downloads and uses KEGG data. Non-academic uses may require a KEGG
## license agreement (details at http://www.kegg.jp/kegg/legal.html).
## ##############################################################################
```

We have 390 metagenome samples collected from underwater cultural heritage sites including plane wrecks and shipwrecks. A few of our samples did not sequence well and have very low trimmed read counts. We will remove those below 7 million reads before further analyzing mapping results.

```
#determine samples to keep
read_counts <- read.csv("data/read_counts.csv",header=F)
colnames(read_counts) <- c("sample","read_count")
ggplot(read_counts,aes(x=read_count))+geom_histogram(bins=100)+geom_vline(xintercept=7e6,color="red")+ggtitle("Pa
ired fastq.gz read count for 390 Maritime eDNA samples")+theme(plot.title=element_text(hjust=0.5))
```



The file minimap2_coverage_and_abundance.csv contains the number of reads mapped for each sample-bin pair using minimap2 to a dRep dereplicated set. We have a set of samples associated with great lakes shipwrecks, but for this project we will focus on submerged aircrafts. Below, total_reads_mapped_per_base is a measurement of depth normalized by the total bin length and bins are of the form Testbed.Site.Source.binnumber.

```
metadata <- read.csv("data/all_metadata.csv")
abundance <- read.csv("data/minimap2_coverage_and_abundance.csv")
abundance <- abundance[abundance$run %in% read_counts[read_counts$read_count>7e6,"sample"],]
abundance <- abundance[!abundance$run %in% metadata[metadata$Testbed=="GreatLakes","SampleID"],]
head(abundance)
```

```
##            run                              bin contig_count total_length
## 1 eDNA-2022-01 GreatLakes.Corsican.Sediment.005         1058      3442476
## 2 eDNA-2022-01 GreatLakes.Corsican.Sediment.006           89       691063
## 3 eDNA-2022-01 GreatLakes.Corsican.Sediment.007          631      1396577
## 4 eDNA-2022-01 GreatLakes.Corsican.Sediment.008            3       949181
## 5 eDNA-2022-01 GreatLakes.Corsican.Sediment.009           74       857965
## 6 eDNA-2022-01 GreatLakes.Corsican.Sediment.010           14       970546
##   total_reads_mapped reads_mapped_per_base   coverage
## 1                501          1.455348e-04 0.24040836
## 2                 48          6.945821e-05 0.09087449
## 3               1074          7.690231e-04 0.17306600
## 4                252          2.654920e-04 0.22271832
## 5                 12          1.398658e-05 0.02925527
## 6                 77          7.933679e-05 0.13950910
```

We also have taxonomic classifications produced by the genome taxonomy database classifier that we will join with the mapping results.

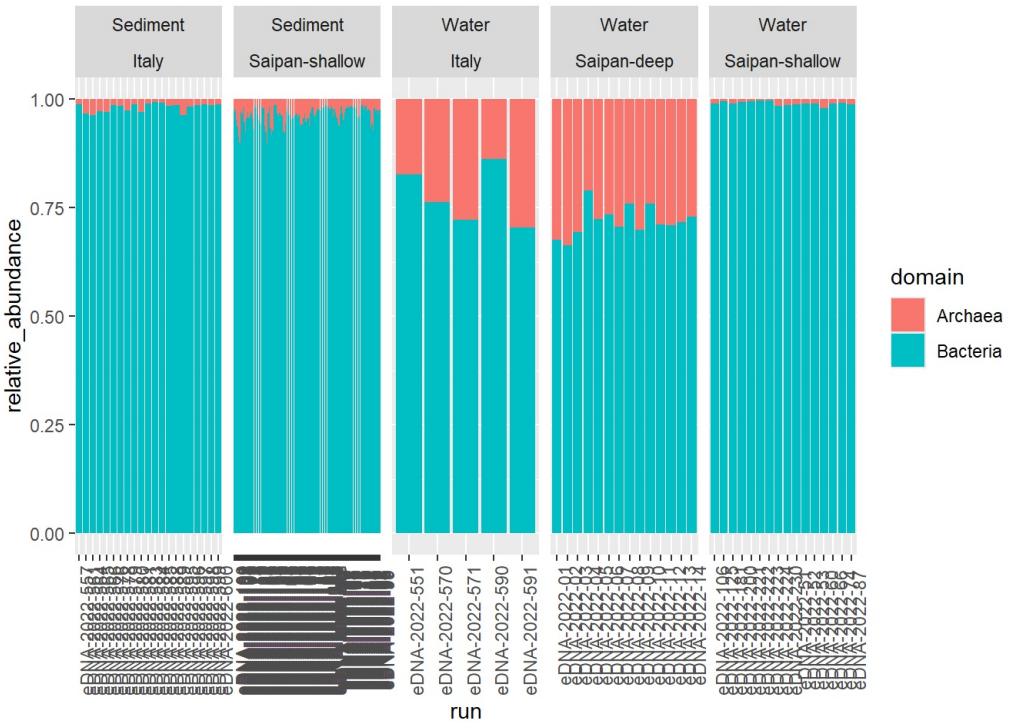
```
gtdbtk <- read.csv("data/gtdbtk.tsv",sep="\t")
gtdbtk <- gtdbtk[,c("bin","classification")]
gtdbtk$domain <- str_replace(str_split(gtdbtk$classification,";",simplify=TRUE)[,1],"d__","")
gtdbtk$phylum <- str_replace(str_split(gtdbtk$classification,";",simplify=TRUE)[,2],"p__","")
gtdbtk$class <- str_replace(str_split(gtdbtk$classification,";",simplify=TRUE)[,3],"c__","")
gtdbtk$order <- str_replace(str_split(gtdbtk$classification,";",simplify=TRUE)[,4],"o__","")
gtdbtk$family <- str_replace(str_split(gtdbtk$classification,";",simplify=TRUE)[,5],"f__","")
gtdbtk$genus <- str_replace(str_split(gtdbtk$classification,";",simplify=TRUE)[,6],"g__","")
gtdbtk$species <- str_replace(str_split(gtdbtk$classification,";",simplify=TRUE)[,7],"s__","")
gtdbtk <- gtdbtk[,colnames(gtdbtk)!="classification"]
head(gtdbtk)
```

```
##                                bin  domain           phylum           class
## 1 GreatLakes.Corsican.Sediment.001 Archaea   Halobacteriota Methanomicrobia
## 2 GreatLakes.Corsican.Sediment.002 Archaea Thermoplasmatota  Thermoplasmata
## 3   GreatLakes.Norman.Sediment.001 Archaea    Nanoarchaeota    Nanoarchaeia
## 4   GreatLakes.Norman.Sediment.002 Archaea Thermoplasmatota  Thermoplasmata
## 5  GreatLakes.Pewabic.Sediment.001 Archaea   Halobacteriota Methanosarcinia
## 6  GreatLakes.Pewabic.Sediment.002 Archaea   Thermoproteota Nitrososphaeria
##                order              family           genus species
## 1 Methanomicrobiales   Methanoregulaceae   Methanoregula
## 2       RBG-16-68-12        RBG-16-68-12    RBG-16-68-12
## 3    Woesearchaeales               ARS49
## 4       RBG-16-68-12        RBG-16-68-12    RBG-16-68-12
## 5  Methanosarcinales Methanoperedenaceae Methanoperedens
## 6  Nitrososphaerales   Nitrosopumilaceae         UBA8516
```

```
abundance <- abundance[,c("run","bin","reads_mapped_per_base","coverage")]
abundance <- join(abundance,gtdbtk,by="bin")
```

To calculate relative abundance, we calculate the total reads_mapped_per_base for each sample, then normalize mapping depths by this value.

```
total_reads_mapped_per_base <- data.frame(abundance%>%group_by(run)%>%summarize(total_reads_mapped_per_base=sum(r
eads_mapped_per_base)))
abundance <- join(abundance,total_reads_mapped_per_base,by="run")
abundance$relative_abundance <- abundance$reads_mapped_per_base/abundance$total_reads_mapped_per_base
total_abundance <- data.frame(abundance%>%group_by(run)%>%summarize(total_abundance=sum(relative_abundance)))
abundance <- abundance[abundance$run %in% total_abundance[!is.na(total_abundance$total_abundance),"run"],]
abundance$SampleID <- abundance$run
abundance <- join(abundance,metadata,by="SampleID")
ggplot(abundance,aes(y=relative_abundance,x=run,fill=domain))+geom_bar(stat="identity")+theme(axis.text.x=element
_text(angle=90))+facet_grid(~Source+Testbed,scales="free")
```

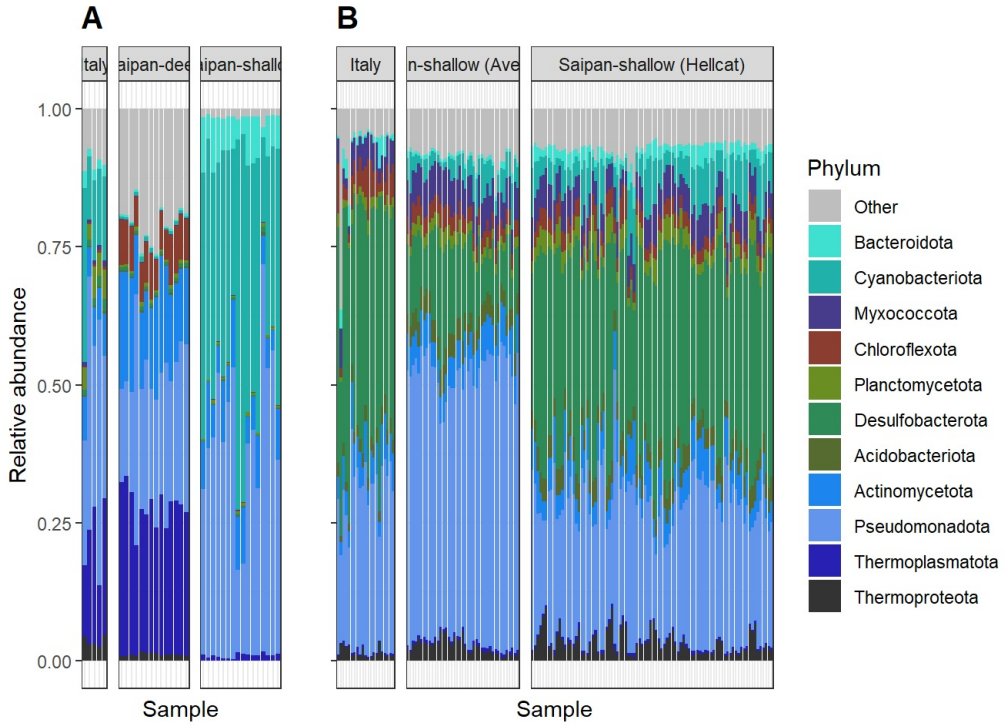


Above we are just visualizing the relative proportions of bacteria and archaea in our samples. We ultimately want to color relative abundances by phylum, but by default there are too many unique phyla to get a clear visualization, so we will group less common phyla into the group “Other”. We will also decide on standard colours to use for each phylum.

```
abundance[grepl("Desulfobacterota",abundance$phylum),"phylum"] <- "Desulfobacterota"
abundance[grepl("Myxococcota",abundance$phylum),"phylum"] <- "Myxococcota"


phylum_abundances <- data.frame(abundance%>%group_by(phylum)%>%summarize(total_abundance=sum(relative_abundance))
)
phylum_abundances <- phylum_abundances[order(phylum_abundances$total_abundance,decreasing = TRUE),]
abundance$temp_phylum <- abundance$phylum
abundance[!abundance$temp_phylum %in% phylum_abundances[1:11,"phylum"],"temp_phylum"] <- "Other"


abundance$temp_phylum <- factor(abundance$temp_phylum,levels=c("Other","Bacteroidota","Cyanobacteriota","Myxococc
ota","Chloroflexota","Planctomycetota","Desulfobacterota","Acidobacteriota","Actinomycetota","Pseudomonadota","Th
ermoplasmatota","Thermoproteota"))

abundance$temp_testbed <- abundance$Testbed
abundance[abundance$Site=="Hellcat","temp_testbed"] <- "Saipan-shallow (Hellcat)"
abundance[abundance$Site=="Avenger","temp_testbed"] <- "Saipan-shallow (Avenger)"
abundance[abundance$Source=="Water" & abundance$Testbed=="Saipan-shallow","temp_testbed"] <- "Saipan-shallow"

phylum_colors <- c(Actinomycetota="dodgerblue2", Pseudomonadota="cornflowerblue",SAR324="royalblue4", Patescibact
eria="khaki4",Cyanobacteriota="lightseagreen",Bacteroidota="turquoise",Nitrospirota="darkorange3",Planctomycetota
="olivedrab",Other="gray",Gemmatimonadota="purple",`Thermoproteota`="gray20",Myxococcota="slateblue4",Methylomira
bilota="brown",Thermoplasmatota="#2720b3",Chloroflexota="coral4","Acidobacteriota"="darkolivegreen",Desulfobacter
ota="seagreen4")
```

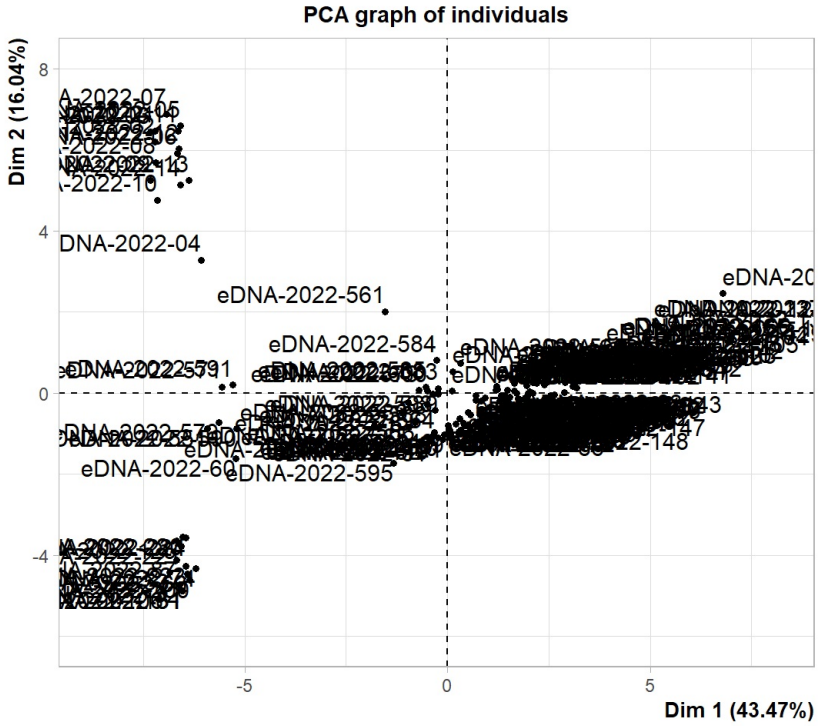
We can now make the first manuscript plot visualizing phylum relative abundances (unintuitively, figure 3).

```
figure_3_ab <- plot_grid(ggplot(abundance[abundance$Source=="Water",],aes(y=relative_abundance,x=run,fill=temp_ph
ylum))+geom_bar(stat="identity")+theme(axis.text.x=element_text(angle=90))+facet_grid(~temp_testbed,scales="free"
,space="free")+scale_fill_manual(values = phylum_colors)+theme_bw()+theme(axis.text.x=element_blank(),axis.ticks.
x = element_blank())+xlab("")+ylab("Relative abundance")+guides(fill=guide_legend(title="Phylum"))+ theme(legend.
position="none")+ggtitle("Water")+xlab("Sample")+ggtitle("A")+ theme(plot.title = element_text(face="bold", size
= 15)),
          ggplot(abundance[abundance$Source=="Sediment",],aes(y=relative_abundance,x=run,fill=temp_phylum))+geom_
bar(stat="identity")+theme(axis.text.x=element_text(angle=90))+facet_grid(~temp_testbed,scales="free",space="free
")+scale_fill_manual(values = phylum_colors)+theme_bw()+theme(axis.text.x=element_blank(),axis.ticks.x = element_
blank())+xlab("")+ylab("")+xlab("Sample")+theme(axis.text.y=element_blank())+guides(fill=guide_legend(title="Phyl
um"))+ggtitle("B")+ theme(plot.title = element_text(face="bold", size = 15)),rel_widths=c(0.4,1))

ggsave("plots/Figure_3_AB.png",width=10.5,height=4,figure_3_ab)


figure_3_ab
```

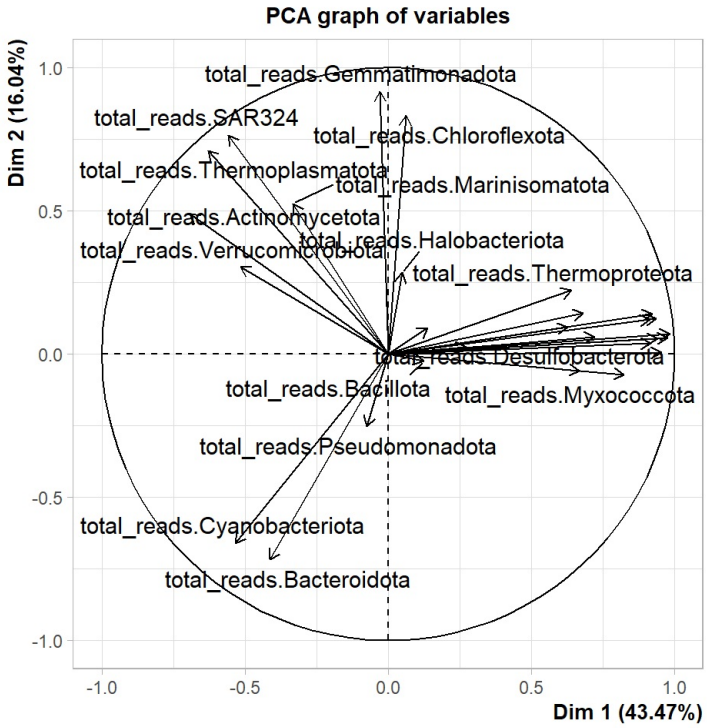


We will first make a pca using the abundances of each unique phylum in each sample as input.

```
abundance$key <- paste(abundance$run,abundance$phylum)
phylum_abundances <- data.frame(abundance%>%group_by(key)%>%summarize(total_reads=sum(relative_abundance)))
phylum_abundances$run <- str_split(phylum_abundances$key," ",simplify=TRUE)[,1]
phylum_abundances$phylum <- str_split(phylum_abundances$key," ",simplify=TRUE)[,2]

pca <- data.frame(reshape(phylum_abundances[,c("total_reads","run","phylum")], idvar = "run", timevar = "phylum",
direction = "wide"))
rownames(pca) <- pca$run
pca <- pca[,colnames(pca)!="run"]
colnames(pca) <- str_replace(colnames(pca),"relative_abundance.","")
pca <- PCA(pca,scale.unit=TRUE,ncp=5,graph=TRUE)
```

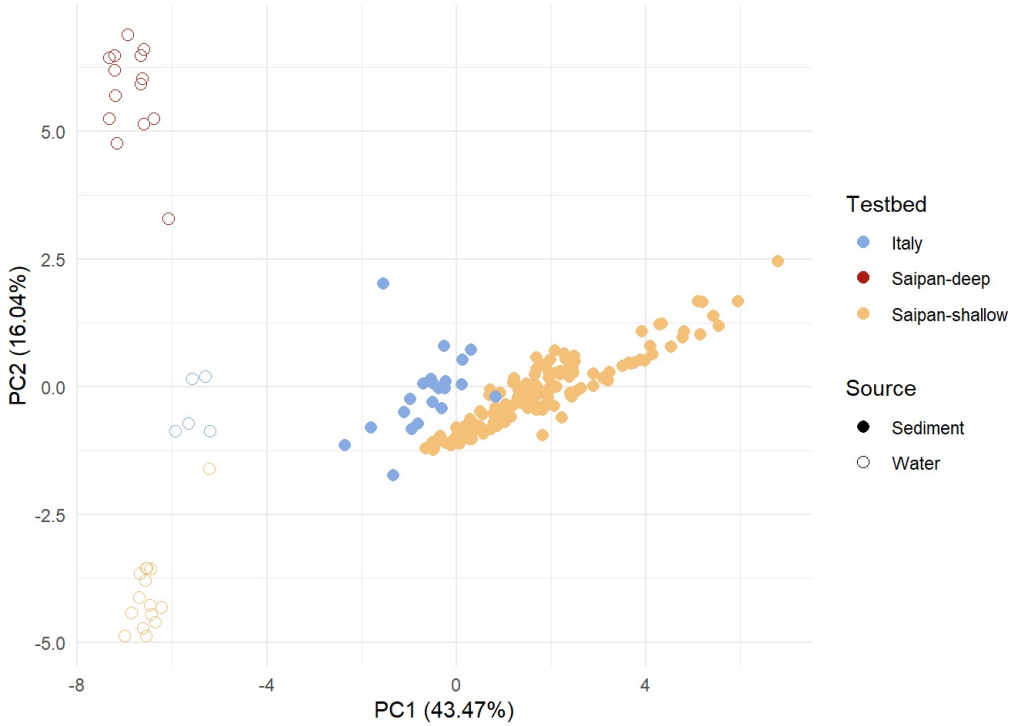



```
## Warning: ggrepel: 13 unlabeled data points (too many overlaps). Consider
## increasing max.overlaps
```



Inspecting different metadata variables suggests that Testbed i.e. location and sample type are key determinants of clustering.

```
pca_data <- data.frame(pca$ind$coord)
pca_data$SampleID <- rownames(pca_data)
pca_data <- join(pca_data,metadata,by="SampleID")

testbed_colors=c(Italy="#88abe3",`Saipan-shallow`="#f5c178",`Saipan-deep`="#ab1c15")

figure_2a <- ggplot(pca_data,aes(x=Dim.1,y=Dim.2,col=Testbed,shape=Source))+geom_point(size=2.5)+scale_shape_manu
al(values=c(16,1))+scale_color_manual(values=testbed_colors)+xlab(paste0("PC1 (",as.character(round(pca$eig[1,"pe
rcentage of variance"],2)),"%)"))+ylab(paste0("PC2 (",as.character(round(pca$eig[2,"percentage of variance"],2)),
"%)"))+theme_minimal()

figure_2a
```



Since we have the phylum abundances calculated, let's run some anosim comparisons using the R package vegan.

```
phylum_abundances <- data.frame(reshape(phylum_abundances[,c("total_reads","run","phylum")], idvar = "run", timev
ar = "phylum", direction = "wide"))
rownames(phylum_abundances) <- phylum_abundances$run
phylum_abundances <- phylum_abundances[,colnames(phylum_abundances)!="run"]
colnames(phylum_abundances) <- str_replace(colnames(phylum_abundances),"total_reads.","")
com <- as.matrix(phylum_abundances)
rownames(metadata) <- metadata$SampleID
phylum_abundances <- cbind(phylum_abundances,metadata[rownames(phylum_abundances),])
ano = anosim(com, phylum_abundances$Source, distance = "bray", permutations = 9999)
ano
```

```
##
## Call:
## anosim(x = com, grouping = phylum_abundances$Source, permutations = 9999,      distance = "bray")
## Dissimilarity: bray
##
## ANOSIM statistic R: 0.946
##       Significance: 1e-04
##
## Permutation: free
## Number of permutations: 9999
```

There is a very significant difference between water and sediment samples, as we would expect from intuition and the visualizations.

```
mysimper <- simper(com,phylum_abundances$Source,permutations = 9999)
mysimper
```

```
## cumulative contributions of most influential species:
##
## $Water_Sediment
## Desulfobacterota  Cyanobacteriota   Pseudomonadota Thermoplasmatota
##        0.2476172        0.4248004        0.5474736        0.6632292
##   Actinomycetota
##        0.7150668
```

The separation by sample type is most strongly influenced by Desulfobacterota, followed by Cyanobacteriota and Pseudomonadota. Note that the reported values are cumulative and we must subtract them to get individual phylum contributions.

```
print(as.numeric(mysimper$Water_Sediment$cusum)[1])
```

```
## [1] 0.2476172
```

```
for (val in seq(2,5)){
  print(as.numeric(mysimper$Water_Sediment$cusum)[val]-as.numeric(mysimper$Water_Sediment$cusum)[val-1])
}
```

```
## [1] 0.1771832
## [1] 0.1226732
## [1] 0.1157556
## [1] 0.05183757
```

You might have noticed in the above barplots that the saipan-deep water samples looked very different from the other water samples. Let's quantify that difference:

```
water_phylum_abundances <- phylum_abundances[phylum_abundances$Source=="Water",]
water_phylum_abundances$is_saipan_deep <- water_phylum_abundances$Testbed=="Saipan-deep"
anosim(com[rownames(water_phylum_abundances),], water_phylum_abundances$is_saipan_deep, distance = "bray", permut
ations = 9999)
```

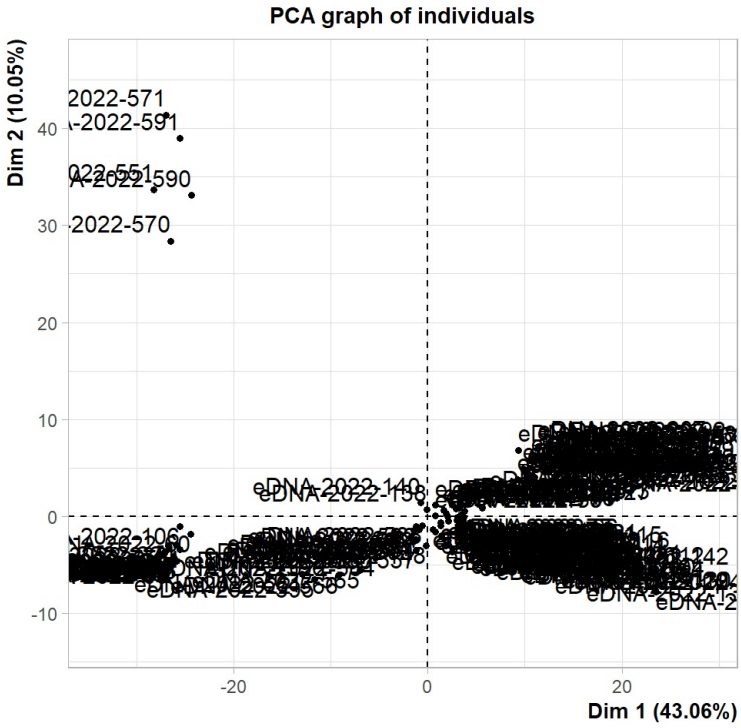
```
##
## Call:
## anosim(x = com[rownames(water_phylum_abundances), ], grouping = water_phylum_abundances$is_saipan_deep,      p
ermutations = 9999, distance = "bray")
## Dissimilarity: bray
##
## ANOSIM statistic R: 0.9083
##       Significance: 1e-04
##
## Permutation: free
## Number of permutations: 9999
```

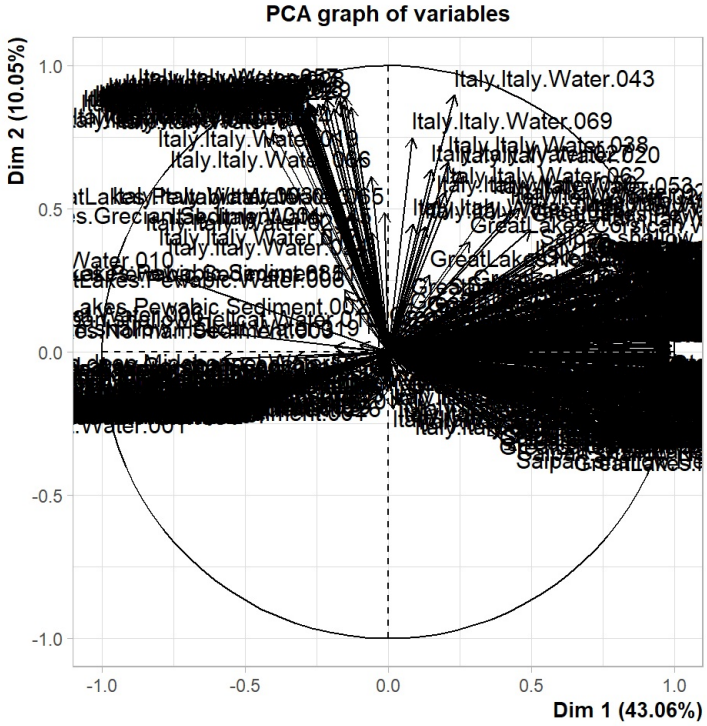
```
simper(com[rownames(water_phylum_abundances),], water_phylum_abundances$is_saipan_deep, permutations = 9999)
```

```
## cumulative contributions of most influential species:
##
## $TRUE_FALSE
##  Cyanobacteriota Thermoplasmatota   Pseudomonadota           SAR324
##        0.2989831        0.4832638        0.6285192        0.7241699
```

Let's see how the pca visualization changes when individual MAGs, not phyla, are used as input.

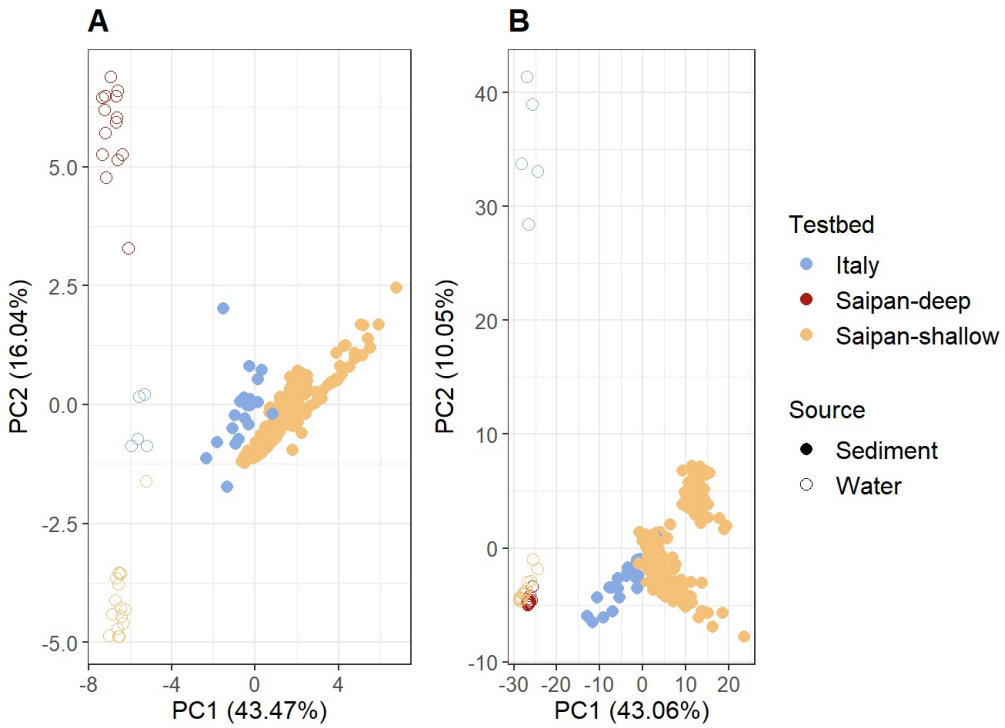
```
pca <- data.frame(reshape(abundance[,c("relative_abundance","run","bin")], idvar = "run", timevar = "bin", direct
ion = "wide"))
rownames(pca) <- pca$run
pca <- pca[,colnames(pca)!="run"]
colnames(pca) <- str_replace(colnames(pca),"relative_abundance.","")
pca <- PCA(pca,scale.unit=TRUE,ncp=5,graph=TRUE)
```






```
pca_bin_contribs <- data.frame(pca$var$contrib)
```

```
pca_data <- data.frame(pca$ind$coord)
pca_data$SampleID <- rownames(pca_data)
pca_data <- join(pca_data,metadata,by="SampleID")

figure_2b <- ggplot(pca_data,aes(x=Dim.1,y=Dim.2,col=Testbed,shape=Source))+geom_point(size=2.5)+scale_shape_manu
al(values=c(16,1))+scale_color_manual(values=testbed_colors)+xlab(paste0("PC1 (",as.character(round(pca$eig[1,"pe
rcentage of variance"],2)),"%")))+ylab(paste0("PC2 (",as.character(round(pca$eig[2,"percentage of variance"],2)),
"%")))+theme_bw()

figure_2 <- plot_grid(figure_2a+theme_bw()+theme(legend.position="none")+ggtitle("A")+ theme(plot.title = element
_text(face="bold", size = 15),axis.text.x=element_text(size=10),axis.title.x=element_text(size=12),axis.text.y=el
ement_text(size=10),axis.title.y=element_text(size=12)),figure_2b+ggtitle("B")+ theme(plot.title = element_text(f
ace="bold", size = 15),axis.text.x=element_text(size=10),axis.title.x=element_text(size=12),axis.text.y=element_t
ext(size=10),axis.title.y=element_text(size=12),legend.text=element_text(size=12),legend.title=element_text(size=
12)),rel_widths=c(0.7,1))

ggsave("plots/Figure_2.png",figure_2,width=10,height=5,bg="white")
figure_2
```



Using individual MAGs as input results in substantial separation of Italy water bins across the second principal component. The two distinct blobs of Saipan-shallow sediment samples in figure B also correspond to the Hellcat and Avenger sites. We can describe the significance of the Italy from other water samples using ANOSIM:

```
mag_abundances <- data.frame(reshape(abundance[,c("relative_abundance","run","bin")], idvar = "run", timevar = "b
in", direction = "wide"))
rownames(mag_abundances) <- mag_abundances$run
mag_abundances <- mag_abundances[,colnames(mag_abundances)!="run"]
colnames(mag_abundances) <- str_replace(colnames(mag_abundances),"relative_abundance.","")
com <- as.matrix(mag_abundances)
rownames(metadata) <- metadata$SampleID
mag_abundances <- cbind(mag_abundances,metadata[rownames(mag_abundances),])
mag_abundances$is_odd <- mag_abundances$Source=="Water" & mag_abundances$Testbed=="Italy"
mag_abundances[mag_abundances$is_odd,"is_odd"] <- "Italy_water"

mag_abundances$is_Italy <- mag_abundances$Testbed=="Italy"
anosim(com[rownames(mag_abundances[mag_abundances$Source=="Water",]),], mag_abundances[mag_abundances$Source=="Wa
ter","is_Italy"], distance = "bray", permutations = 9999)
```

```
##
## Call:
## anosim(x = com[rownames(mag_abundances[mag_abundances$Source ==      "Water", ]), ], grouping = mag_abundances
[mag_abundances$Source ==      "Water", "is_Italy"], permutations = 9999, distance = "bray")
## Dissimilarity: bray
##
## ANOSIM statistic R: 0.3932
##       Significance: 1e-04
##
## Permutation: free
## Number of permutations: 9999
```

There is a significant difference between Italy and other water samples. However, this significance disappears if the phylum abundances, rather than MAG abundances, are used as input:

```
for (col in colnames(phylum_abundances)){
  if (col %in% gtdbtk$phylum){
    phylum_abundances[,col] <- as.numeric(phylum_abundances[,col])
  }
}
com <- as.matrix(phylum_abundances[,!colnames(phylum_abundances) %in% colnames(metadata) & colnames(phylum_abunda
nces)!="is_Italy"])
phylum_abundances$is_Italy <- phylum_abundances$Testbed=="Italy"
anosim(com[rownames(phylum_abundances[phylum_abundances$Source=="Water",]),], phylum_abundances[phylum_abundances
$Source=="Water","is_Italy"], distance = "bray", permutations = 9999)
```

```
##
## Call:
## anosim(x = com[rownames(phylum_abundances[phylum_abundances$Source ==      "Water", ]), ], grouping = phylum_a
bundances[phylum_abundances$Source ==      "Water", "is_Italy"], permutations = 9999, distance = "bray")
## Dissimilarity: bray
##
## ANOSIM statistic R: -0.02742
##       Significance: 0.647
##
## Permutation: free
## Number of permutations: 9999
```

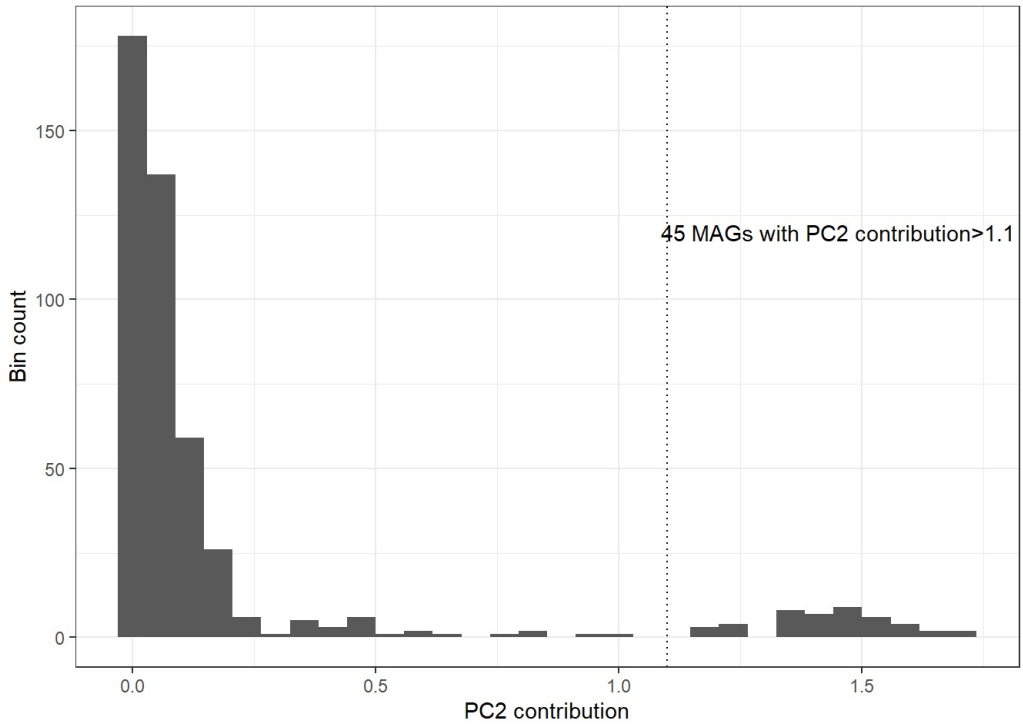
Let's determine which specific MAGs are leading to this difference by leveraging the second principal component where the Italy samples are separated.

```
figure_s2 <- ggplot(pca_bin_contribs,aes(x=Dim.2))+geom_histogram()+geom_vline(xintercept=1.1,linetype="dotted")+
theme_bw()+xlab("PC2 contribution")+ylab("Bin count") +annotate("text", x = 1.45, y = 120, label = "45 MAGs with
PC2 contribution>1.1")
ggsave("plots/Figure_s2.png",figure_s2,height=5,width=8)
```

```
## `stat_bin()` using `bins = 30`. Pick better value with `binwidth`.
```

```
figure_s2
```

```
## `stat_bin()` using `bins = 30`. Pick better value with `binwidth`.
```



The top peak of PC2 contributions contains 45 bins:

```
odd_bins <- rownames(pca_bin_contribs[pca_bin_contribs$Dim.2>1.1,])
length(odd_bins)
```

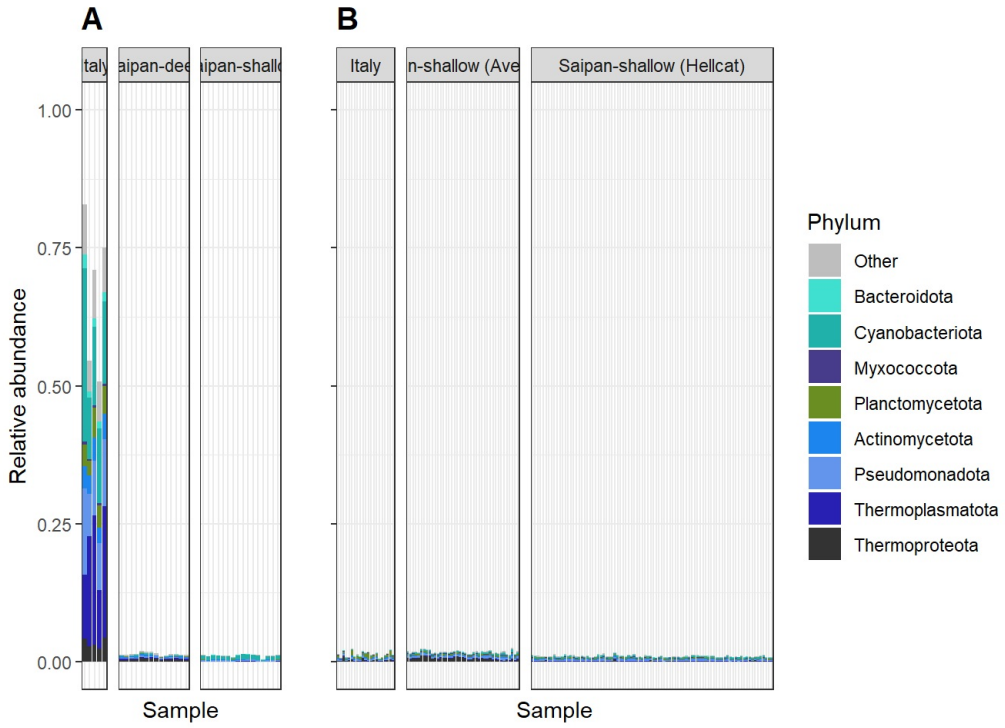
```
## [1] 45
```

```
odd_bins
```

```
##  [1] "Italy.Italy.Water.015" "Italy.Italy.Water.001" "Italy.Italy.Water.022"
##  [4] "Italy.Italy.Water.025" "Italy.Italy.Water.028" "Italy.Italy.Water.029"
##  [7] "Italy.Italy.Water.002" "Italy.Italy.Water.030" "Italy.Italy.Water.031"
## [10] "Italy.Italy.Water.004" "Italy.Italy.Water.005" "Italy.Italy.Water.032"
## [13] "Italy.Italy.Water.006" "Italy.Italy.Water.034" "Italy.Italy.Water.035"
## [16] "Italy.Italy.Water.036" "Italy.Italy.Water.008" "Italy.Italy.Water.007"
## [19] "Italy.Italy.Water.037" "Italy.Italy.Water.040" "Italy.Italy.Water.041"
## [22] "Italy.Italy.Water.043" "Italy.Italy.Water.044" "Italy.Italy.Water.046"
## [25] "Italy.Italy.Water.047" "Italy.Italy.Water.048" "Italy.Italy.Water.009"
## [28] "Italy.Italy.Water.050" "Italy.Italy.Water.051" "Italy.Italy.Water.049"
## [31] "Italy.Italy.Water.010" "Italy.Italy.Water.011" "Italy.Italy.Water.054"
## [34] "Italy.Italy.Water.055" "Italy.Italy.Water.052" "Italy.Italy.Water.057"
## [37] "Italy.Italy.Water.059" "Italy.Italy.Water.060" "Italy.Italy.Water.012"
## [40] "Italy.Italy.Water.013" "Italy.Italy.Water.064" "Italy.Italy.Water.014"
## [43] "Italy.Italy.Water.067" "Italy.Italy.Water.068" "Italy.Italy.Water.069"
```

We can produce a quick visualization of these bins' relative abundance to add to figure 3:

```
figure_3_cd <- plot_grid(ggplot(abundance[abundance$Source=="Water" & abundance$bin %in% odd_bins,],aes(y=relativ
e_abundance,x=run,fill=temp_phylum))+theme_bw()+ylim(0,1)+geom_bar(stat="identity")+theme(axis.text.x=element_tex
t(angle=90))+facet_grid(~temp_testbed,scales="free",space="free")+scale_fill_manual(values = phylum_colors)+theme
(axis.text.x=element_blank(),axis.ticks.x = element_blank())+xlab("")+ylab("Relative abundance")+guides(fill=guid
e_legend(title="Phylum"))+ theme(legend.position="none")+ggtitle("Water")+xlab("Sample")+ggtitle("A")+ theme(plot
.title = element_text(face="bold", size = 15)),
          ggplot(abundance[abundance$Source=="Sediment" &  abundance$bin %in% odd_bins,],aes(y=relative_abundance
,x=run,fill=temp_phylum))+geom_bar(stat="identity")+theme_bw()+theme(axis.text.x=element_text(angle=90))+facet_gr
id(~temp_testbed,scales="free",space="free")+ylim(0,1)+scale_fill_manual(values = phylum_colors)+theme(axis.text.
x=element_blank(),axis.ticks.x = element_blank())+xlab("")+ylab("")+xlab("Sample")+theme(axis.text.y=element_blan
k())+guides(fill=guide_legend(title="Phylum"))+ggtitle("B")+ theme(plot.title = element_text(face="bold", size =
15)),rel_widths=c(0.4,1))
ggsave("plots/Figure_3_CD.png",width=10.5,height=4,figure_3_cd)
figure_3_cd
```



We can use simple fisher tests to compare the eggnog coding potential of these MAGs to the rest of the dataset. This requires a bit of manipulation to get the eggnog kegg annotations into a suitable format for enrichment analysis.

```
#Using an excel converted version of the file because R sometimes has trouble reading in very long tsvs
eggnog <- read_xlsx("data/emapper.annotations.xlsx")
eggnog <- data.frame(eggnog)
eggnog$bin <- str_split(eggnog$query,"_",simplify=TRUE)[,1]
#get a matrix of how often each kegg id occurs in each MAG
kegg <- eggnog[,c("query","KEGG_ko")]
kegg <- kegg[kegg$KEGG_ko!="-",]
kegg <- kegg[!is.na(kegg$KEGG_ko),]
kegg <- kegg[kegg$KEGG_ko!="",]
kegg <- data.table(kegg)
kegg <- kegg[, list(kegg_hit = unlist(strsplit(KEGG_ko , ","))),by=query]
kegg <- data.frame(kegg)

kegg$group <- paste(str_split(kegg$query,fixed("."),simplify=TRUE)[,1],str_split(kegg$query,fixed("."),simplify=T
RUE)[,2],str_split(kegg$query,fixed("."),simplify=TRUE)[,3],sep=".")
kegg$bin <- str_split(kegg$query,"_",simplify=TRUE)[,1]
kegg <- distinct(kegg[,c("bin","kegg_hit")])
colnames(kegg) <- c("fasta","kegg_hit")
kegg$key <- paste(kegg$fasta,kegg$kegg_hit,sep=";")
kegg_counts <- data.frame(table(kegg$key))
kegg_counts$bin <- str_split(kegg_counts$Var1,";",simplify=TRUE)[,1]
kegg_counts$kegg <- str_split(kegg_counts$Var1,";",simplify=TRUE)[,2]
temp <- data.frame(table(kegg_counts$kegg))
kegg_counts <- kegg_counts[kegg_counts$kegg %in% temp[temp$Freq>5,"Var1"],]
kegg_counts <- data.frame(reshape(kegg_counts[,c("bin","kegg","Freq")],idvar = "bin", timevar = "kegg", direction
= "wide"))
rownames(kegg_counts) <- kegg_counts$bin
kegg_counts <- kegg_counts[,colnames(kegg_counts)!="bin"]
colnames(kegg_counts) <- str_replace_all(colnames(kegg_counts),"Freq.","")
kegg_counts[is.na(kegg_counts)] <- 0

kegg_counts <- data.frame(t(kegg_counts))
```

```
#get matrix of how often each kegg id occurs in each clade (Italy or other)
clades <- list()
clades[["Italy_75"]] <- odd_bins
clades[["outgroup"]] <- colnames(kegg_counts)[!colnames(kegg_counts) %in% clades[["Italy_75"]]]


all_counts <- list()
for (clade in names(clades)){

  clade_counts <- kegg_counts[,clades[[clade]]]
  clade_counts$sum <- apply(clade_counts,1,sum)
  clade_counts$bins_with_kegg <- ncol(clade_counts)-1-clade_counts$sum
  all_counts[[clade]] <- clade_counts[,c("sum","bins_with_kegg")]
  all_counts[[clade]]$kegg <- rownames(clade_counts)

}
all_counts <- ldply(all_counts)
colnames(all_counts) <- c("bin_group","bins_with_kegg","bins_without_kegg","kegg")
```

Now we can run fisher tests.

```
all_fisher_results <- list()

get_fold_change <- function(x){
  x <- as.numeric(x)
  return(x[1]/(x[1]+x[2])/(x[3]/(x[3]+x[4])))

}

for (clade in names(clades)){

  within_clade_counts <- all_counts[all_counts$bin_group==clade,]
  outside_clade_counts <- all_counts[all_counts$bin_group!=clade,]

  rownames(outside_clade_counts) <- outside_clade_counts$kegg
  rownames(within_clade_counts) <- within_clade_counts$kegg
  outside_clade_counts <- outside_clade_counts[rownames(within_clade_counts),]
  clade_counts <- cbind(within_clade_counts[,c("bins_with_kegg","bins_without_kegg")],outside_clade_counts[,c("bi
ns_with_kegg","bins_without_kegg")])
  colnames(clade_counts) <- c("ingroup_with_kegg","ingroup_without_kegg","outgroup_with_kegg","outgroup_without_k
egg")
p_values <- apply(clade_counts,1, function(x) fisher.test(matrix(x,nr=2))$p.value)
fold_changes <- log2(apply(clade_counts,1,get_fold_change))
#fisher_results <- fisher_results[[clade]]
fisher_results <- clade_counts
fisher_results$fold_change <- fold_changes
fisher_results$p_value <- p_values
#fisher_results <- data.frame(rownames(within_clade_counts),fold_changes,p_values)
#colnames(fisher_results) <- c("orthogroup","fold_change","p_value")
fisher_results[is.nan(fisher_results$fold_change),"fold_change"] <- 0
fisher_results[is.infinite(fisher_results$fold_change) & fisher_results$fold_change < 0,"fold_change"] <- min(fis
her_results[!is.infinite(fisher_results$fold_change),"fold_change"],na.rm = TRUE)
fisher_results[is.infinite(fisher_results$fold_change) & fisher_results$fold_change > 0,"fold_change"] <- max(fis
her_results[!is.infinite(fisher_results$fold_change),"fold_change"],na.rm = TRUE)
fisher_results$p_adjust <- p.adjust(fisher_results$p_value, method = 'hochberg', n = nrow(fisher_results))
 fisher_results <- fisher_results[order(fisher_results$p_value),]
 fisher_results$kegg <- rownames(fisher_results)
all_fisher_results[[clade]] <- fisher_results
print(clade)

  }
```

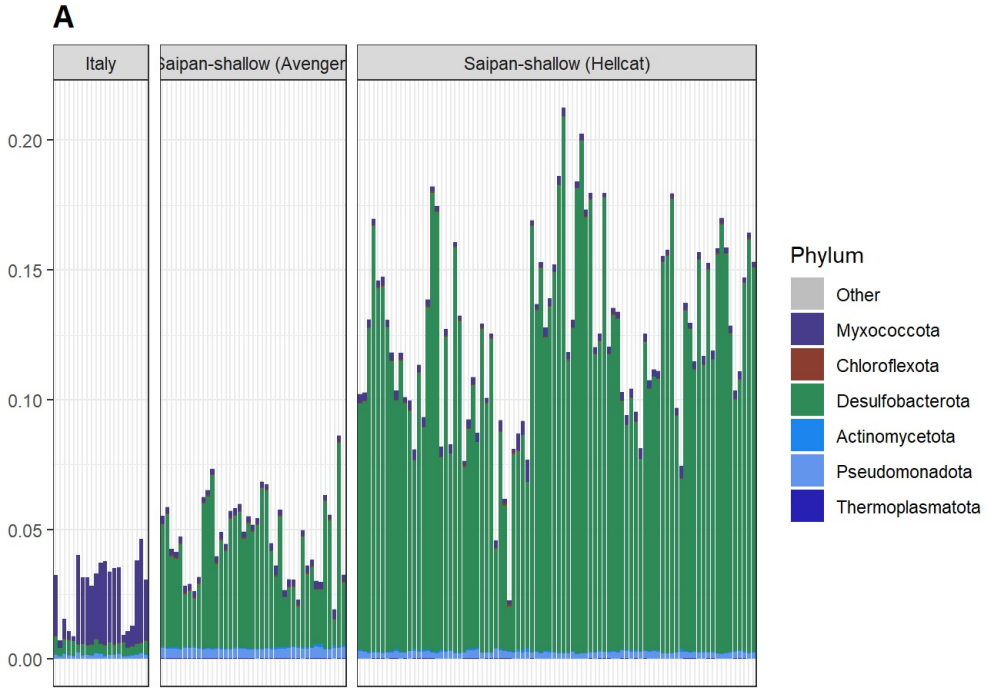
```
## [1] "Italy_75"
## [1] "outgroup"
```

```
all_fisher_results <- ldply(all_fisher_results)
all_fisher_results <- all_fisher_results[all_fisher_results$.id=="Italy_75",]
all_fisher_results <- all_fisher_results[all_fisher_results$p_adjust<=0.05,]
all_fisher_results <- all_fisher_results[order(all_fisher_results$fold_change),]
```

```
kegg_names <- read.table("data/ko00001.keg",fill=TRUE)
kegg_names <- kegg_names[kegg_names$V1 %in% c("A","B","C","D"),]
kegg_names$name <- paste(kegg_names$V3,kegg_names$V4,kegg_names$V5,kegg_names$V6)
kegg_names$kegg <- kegg_names$V2
kegg_names <- kegg_names[,c("kegg","name")]
all_fisher_results$kegg <- str_replace(all_fisher_results$kegg,"ko.","")
all_fisher_results <- plyr::join(all_fisher_results,kegg_names,by="kegg")

all_fisher_results <- distinct(all_fisher_results)
all_fisher_results <- all_fisher_results[order(all_fisher_results$p_value),]
write.csv(all_fisher_results,"data/italy_fisher_results.csv")
```

Let's move on to alkane metabolism occurring in sediment sites. We will first visualize the abundance of bins encoding alkane 1-monooxygenase, a key enzyme in converting alkanes into a metabolizable form.

```
figure_4a <- ggplot(abundance[abundance$Source=="Sediment" & abundance$bin %in% eggnog[grepl("Alkane 1-monooxygen
ase",eggnog$Description),"bin"],],aes(y=relative_abundance,x=run,fill=temp_phylum))+
  geom_bar(stat="identity")+theme(axis.text.x=element_text(angle=90))+facet_grid(~temp_testbed,scales="free",spac
e="free")+
  theme_bw()+scale_fill_manual(values = phylum_colors)+theme(axis.text.x=element_blank(),axis.ticks.x = element_b
lank())+xlab("")+ylab("")+ggtitle("A")+ theme(plot.title = element_text(face="bold", size = 15))+guides(fill=guid
e_legend(title="Phylum"))
ggsave("plots/Figure_4a.png",figure_4a+ylab("Relative abundance"),height=3,width=8,dpi=500)
figure_4a
```



Desulfobacterota bins encoding this enzyme appear quite abundant at the Hellcat site. For visualizing the full alkane degradation pathway, I am reading in a list of relevant kos that we will visualize for desulfobacterota at the hellcat site using pathview.

```
to_join <- data.frame(eggnog[,c("bin","KEGG_ko")]%>%separate_longer_delim(delim=",",cols="KEGG_ko"))

target_kos <- read.csv("data/fatty_acid_kos.txt.txt",header=F)

grouped_hellcat_abundances <- join(abundance[abundance$Site=="Hellcat" & abundance$Source=="Sediment" & abundance
$phylum=="Desulfobacterota",],to_join[to_join$KEGG_ko %in% target_kos$V1,c("bin","KEGG_ko")],by="bin")
grouped_hellcat_abundances <- distinct(grouped_hellcat_abundances[,c("run","bin","KEGG_ko","relative_abundance")]
)
grouped_hellcat_abundances <- data.frame(grouped_hellcat_abundances%>%group_by(KEGG_ko)%>%summarize(total_abundan
ce=sum(relative_abundance)))
grouped_hellcat_abundances$total_abundance <- grouped_hellcat_abundances$total_abundance/nrow(metadata[metadata$S
ite=="Hellcat" & metadata$Source=="Sediment",])#normalize total abundances by total number of hellcat sediment ru
ns to get average abundance of bins containing each ko
grouped_hellcat_abundances <- grouped_hellcat_abundances[!is.na(grouped_hellcat_abundances$KEGG_ko),]
temp <- str_replace(grouped_hellcat_abundances$KEGG_ko,"ko:","")
grouped_hellcat_abundances <- grouped_hellcat_abundances$total_abundance
names(grouped_hellcat_abundances) <- temp


pv.out <- pathview(grouped_hellcat_abundances, pathway.id = "ko00071",species="ko",low="black",mid="white",high="
seagreen4",limit=list(gene=0.30, cpd=0.30))
```

```
## Note: Only KEGG ortholog gene ID is supported, make sure it looks like "K01488"!
```

```
## Note: Mapping via KEGG gene ID (not Entrez) is supported for this species,
## it looks like "K01488"!
```

```
## Info: Downloading xml files for ko00071, 1/1 pathways..
```

```
## Info: Downloading png files for ko00071, 1/1 pathways..
```

```
## Info: Working in directory C:/Users/Rayna/Documents/GitHub/UCH_aircrafts
```

```
## Info: Writing image file ko00071.pathview.png
```

The file ko00071.pathview.png was used to produce figure 4c.